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Java SE 27 & JDK 27

Class MathContext

java.lang.Object
    java.math.MathContext

All Implemented Interfaces:

Serializable

```
public final class MathContext
extends Object
implements Serializable
```

Immutable objects which encapsulate the context settings which describe certain rules for numerical operators, such as those implemented by the BigDecimal class.

The base-independent settings are:

1. `precision`: the number of digits to be used for an operation; results are rounded to this precision
2. `roundingMode`: a RoundingMode object which specifies the algorithm to be used for rounding.

Since:

1.5

External Specifications

IEEE Standard for Floating-Point Arithmetic

See Also:

BigDecimal, RoundingMode, Serialized Form

Field Summary

Fields

Modifier and Type	Field	Description
static final MathContext	**DECIMAL128**	A MathContext object with a precision setting matching the precision of the IEEE 754-2019 decimal128 format, 34 digits, and a rounding mode of HALF_EVEN.
static final MathContext	**DECIMAL32**	A MathContext object with a precision setting matching the precision

		of the IEEE 754-2019 decimal32 format, 7 digits, and a rounding mode of HALF_EVEN.
static final MathContext	DECIMAL64	A MathContext object with a precision setting matching the precision of the IEEE 754-2019 decimal64 format, 16 digits, and a rounding mode of HALF_EVEN.
static final MathContext	UNLIMITED	A MathContext object whose settings have the values required for unlimited precision arithmetic.

Constructor Summary

Constructors

Constructor	Description
MathContext(int setPrecision)	Constructs a new MathContext with the specified precision and the HALF_UP rounding mode.
MathContext(int setPrecision, RoundingMode setRoundingMode)	Constructs a new MathContext with a specified precision and rounding mode.
MathContext(String val)	Constructs a new MathContext from a string.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description

boolean	**equals**(Object x)	Compares this MathContext with the specified Object for equality.
int	**getPrecision**()	Returns the precision setting.
RoundingMode	**getRoundingMode**()	Returns the roundingMode setting.
int	**hashCode**()	Returns the hash code for this MathContext.
String	**toString**()	Returns the string representation of this MathContext.

Methods declared in class Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Field Details

UNLIMITED

```
public static final MathContext UNLIMITED
```

A MathContext object whose settings have the values required for unlimited precision arithmetic. The values of the settings are: precision=0 roundingMode=HALF_UP

DECIMAL32

```
public static final MathContext DECIMAL32
```

A MathContext object with a precision setting matching the precision of the IEEE 754-2019 decimal32 format, 7 digits, and a rounding mode of HALF_EVEN. Note the exponent range of decimal32 (min exponent of -95, max exponent of 96) is *not* used for rounding.

DECIMAL64

```
public static final MathContext DECIMAL64
```

A MathContext object with a precision setting matching the precision of the IEEE 754-2019 decimal64 format, 16 digits, and a rounding mode of HALF_EVEN. Note the exponent range of decimal64 (min exponent of -383, max exponent of 384) is *not* used for rounding.

DECIMAL128

```
public static final MathContext DECIMAL128
```

A MathContext object with a precision setting matching the precision of the IEEE 754-2019 decimal128 format, 34 digits, and a rounding mode of HALF_EVEN. Note the exponent range of decimal128 (min exponent of -6143, max exponent of 6144) is *not* used for rounding.

Constructor Details

MathContext

```
public MathContext(int setPrecision)
```

Constructs a new MathContext with the specified precision and the HALF_UP rounding mode.

Parameters:

setPrecision - The non-negative int precision setting.

Throws:

IllegalArgumentException - if the setPrecision parameter is less than zero.

MathContext

```
public MathContext(int setPrecision,
                   RoundingMode setRoundingMode)
```

Constructs a new MathContext with a specified precision and rounding mode.

Parameters:

setPrecision - The non-negative int precision setting.

setRoundingMode - The rounding mode to use.

Throws:

IllegalArgumentException - if the setPrecision parameter is less than zero.

NullPointerException - if the rounding mode argument is null

MathContext

```
public MathContext(String val)
```

Constructs a new `MathContext` from a string. The string must be in the same format as that produced by the `toString()` method.

An `IllegalArgumentException` is thrown if the precision section of the string is out of range (`< 0`) or the string is not in the format created by the `toString()` method.

Parameters:

`val` - The string to be parsed

Throws:

`IllegalArgumentException` - if the precision section is out of range or of incorrect format

`NullPointerException` - if the argument is `null`

Method Details

getPrecision

```
public int getPrecision()
```

Returns the `precision` setting. This value is always non-negative.

Returns:

an `int` which is the value of the `precision` setting

getRoundingMode

```
public RoundingMode getRoundingMode()
```

Returns the roundingMode setting. This will be one of `RoundingMode.CEILING`, `RoundingMode.DOWN`, `RoundingMode.FLOOR`, `RoundingMode.HALF_DOWN`, `RoundingMode.HALF_EVEN`, `RoundingMode.HALF_UP`, `RoundingMode.UNNECESSARY`, or `RoundingMode.UP`.

Returns:

a `RoundingMode` object which is the value of the `roundingMode` setting

equals

```
public boolean equals(Object x)
```

Compares this `MathContext` with the specified `Object` for equality.

Overrides:

equals in class Object

Parameters:

x - Object to which this MathContext is to be compared.

Returns:

true if and only if the specified Object is a MathContext object which has exactly the same settings as this object

See Also:

Object.hashCode(), HashMap

hashCode

```
public int hashCode()
```

Returns the hash code for this MathContext.

Overrides:

hashCode in class Object

Returns:

hash code for this MathContext

See Also:

Object.equals(java.lang.Object),
System.identityHashCode(Object)

toString

```
public String toString()
```

Returns the string representation of this MathContext. The String returned represents the settings of the MathContext object as two space-delimited words (separated by a single space character, '\u0020', and with no leading or trailing white space), as follows:

1. The string "precision=", immediately followed by the value of the precision setting as a numeric string as if generated by the Integer.toString method.
2. The string "roundingMode=", immediately followed by the value of the roundingMode setting as a word. This word will be the same as the name of the corresponding public constant in the RoundingMode enum.

For example:

```
precision=9 roundingMode=HALF_UP
```

Additional words may be appended to the result of toString in the future if more properties are added to this class.

Overrides:

`toString` in class `Object`

Returns:

a `String` representing the context settings

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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